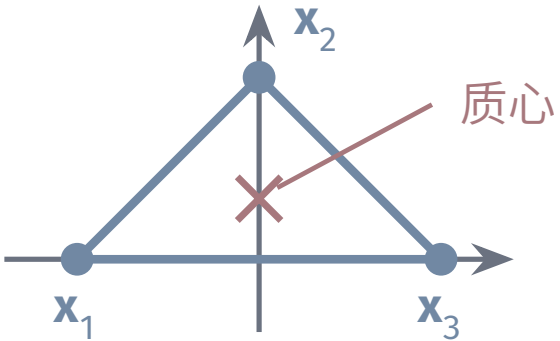


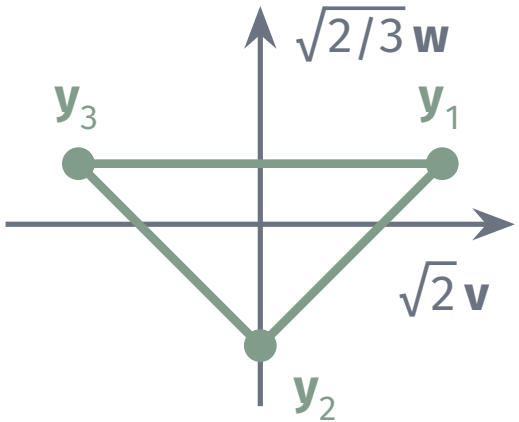
(a)

$$\Delta = \begin{bmatrix} 0 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 0 \end{bmatrix}$$

$$-\frac{1}{2}\mathbf{J} \cdot \mathbf{J}$$

$$\mathbf{B} = \frac{1}{9}\begin{bmatrix} 10 & -2 & -8 \\ -2 & 4 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

(b)



(c)